



LYSO Scintillators Datasheet

Description: Lutetium – Yttrium Oxyorthosilicate, Ce^{3+} doped crystals. LYSO has fast timing property, high Z number and is used for applications where a high stopping power and fast decay time is needed.

Scintillation crystal	: LSO ($\text{Lu}_{1.8}\text{Y}_2\text{SiO}_5:\text{Ce}$)
Density	: 7.2 g/cm^3
Hygroscopic	: No
Wavelength of emission max.	: 420 nm
Refractive index	: 1.82 (420 nm)
Decay time	: 50 ns
Energy resolution	: 8.0%
Light yield	: 30 photons/keV gammas
Photoelectron yield	: 76% relative to NaI(Tl)
Ce concentration	: 0.5 mol %